

The Finite Element Method (FEM) in the study of continuum dynamics - Solid Finite Elements

Introduction

- 1-D elements: beam / truss
- 2-D elements: plates, shells
- 3-D (solid) elements: tetrahedral, hexahedral

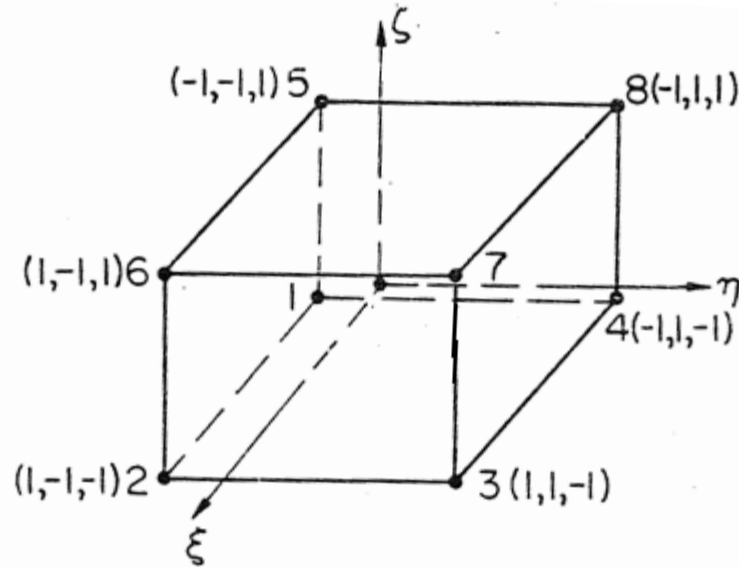
Solid elements:

There is no prevailing dimension in the elastic solid(s) considered

It is not possible to reduce from 3D to 1D/2D but:

Only displacements (no rotations) need to be considered

An example: the hexahedral solid Finite Element with 8 nodes



$$\underline{x}_k = \left\{ \begin{array}{c} u_1 \\ v_1 \\ w_1 \\ \vdots \\ w_8 \end{array} \right\}_{24,1}$$

$$(\xi_1, \eta_1, \zeta_1) = (-1, -1, -1)$$

$$(\xi_2, \eta_2, \zeta_2) = (1, -1, -1)$$

...

$$(\xi_8, \eta_8, \zeta_8) = (-1, 1, 1)$$

Hexahedral element with 8 nodes – shape functions

$$u(\xi, \eta, \zeta) = \sum_{i=1}^8 f_i(\xi, \eta, \zeta) u_i$$

$$v(\xi, \eta, \zeta) = \sum_{i=1}^8 f_i(\xi, \eta, \zeta) v_i$$

$$w(\xi, \eta, \zeta) = \sum_{i=1}^8 f_i(\xi, \eta, \zeta) w_i$$

$$f_i(\xi_j, \eta_j, \zeta_j) = \delta_{ij}$$

$$f_i = \frac{1}{8} (1 + \xi \xi_i)(1 + \eta \eta_i)(1 + \zeta \zeta_i)$$

$$\underline{u} = \begin{Bmatrix} u \\ v \\ w \end{Bmatrix} \quad \underline{u} = [f] \underline{x}_k \quad [f] = \begin{bmatrix} f_1 & 0 & 0 & \cdots & f_8 & 0 & 0 \\ 0 & f_1 & 0 & \cdots & 0 & f_8 & 0 \\ 0 & 0 & f_1 & \cdots & 0 & 0 & f_8 \end{bmatrix}$$

Kinetic Energy – Mass matrix

$$\underline{\dot{u}} = [f] \dot{\underline{x}}_k$$
$$T_k = \frac{1}{2} \int_{Vol} \underline{\dot{u}}^T \rho \underline{\dot{u}} dVol = \frac{1}{2} \dot{\underline{x}}_k^T \underbrace{\int_{Vol} [f]^T \rho [f] dVol}_{[M_k]} \dot{\underline{x}}_k$$

24 x 24 square, symmetric, constant matrix

Elastic potential energy (I)

$$V_k = \frac{1}{2} \int_{Vol} \underline{\varepsilon}^T \underline{\sigma} dVol \quad (\text{Clapeyron})$$

$$\underline{\varepsilon} = \begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \varepsilon_z \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{Bmatrix} \quad \underline{\sigma} = \begin{Bmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \\ \tau_{xy} \\ \tau_{yz} \\ \tau_{zx} \end{Bmatrix}$$

Elastic potential energy (II) – stress-strain relationship

$$\varepsilon_x = \frac{1}{E} \sigma_x - \frac{\nu}{E} (\sigma_y + \sigma_z)$$

$$\varepsilon_y = \frac{1}{E} \sigma_y - \frac{\nu}{E} (\sigma_z + \sigma_x)$$

$$\varepsilon_z = \frac{1}{E} \sigma_z - \frac{\nu}{E} (\sigma_x + \sigma_y)$$

$$\gamma_{xy} = \frac{1}{G} \tau_{xy}$$

$$\gamma_{yz} = \frac{1}{G} \tau_{yz}$$

$$\gamma_{zx} = \frac{1}{G} \tau_{zx}$$



$$\underline{\varepsilon} = [\underline{E}]^{-1} \underline{\sigma}$$



$$\underline{\sigma} = [\underline{E}] \underline{\varepsilon}$$

$$[\underline{E}] = \begin{bmatrix} [E_1] & [0] \\ [0] & [G] \end{bmatrix}$$

$$[E_1] = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & \nu \\ \nu & 1-\nu & \nu \\ \nu & \nu & 1-\nu \end{bmatrix}$$

$$[G] = \begin{bmatrix} G & 0 & 0 \\ 0 & G & 0 \\ 0 & 0 & G \end{bmatrix}$$

Elastic potential energy (III) – strain-displacement relationship

$$\varepsilon_x = \frac{\partial u}{\partial \xi}$$

$$\varepsilon_y = \frac{\partial v}{\partial \eta}$$

$$\varepsilon_z = \frac{\partial w}{\partial \zeta}$$

$$\gamma_{xy} = \frac{\partial u}{\partial \eta} + \frac{\partial v}{\partial \xi}$$

$$\gamma_{yz} = \frac{\partial v}{\partial \zeta} + \frac{\partial w}{\partial \eta}$$

$$\gamma_{zx} = \frac{\partial w}{\partial \xi} + \frac{\partial u}{\partial \zeta}$$

$$\underline{\varepsilon} = \begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \varepsilon_z \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{Bmatrix}$$

$$= [D] \underline{u}$$

$$[D] =$$

$$\begin{bmatrix} \frac{\partial}{\partial \xi} & 0 & 0 \\ 0 & \frac{\partial}{\partial \eta} & 0 \\ 0 & 0 & \frac{\partial}{\partial \zeta} \\ \frac{\partial}{\partial \eta} & \frac{\partial}{\partial \xi} & 0 \\ 0 & \frac{\partial}{\partial \zeta} & \frac{\partial}{\partial \eta} \\ \frac{\partial}{\partial \zeta} & 0 & \frac{\partial}{\partial \xi} \end{bmatrix}$$

Elastic potential energy (IV) – stiffness matrix

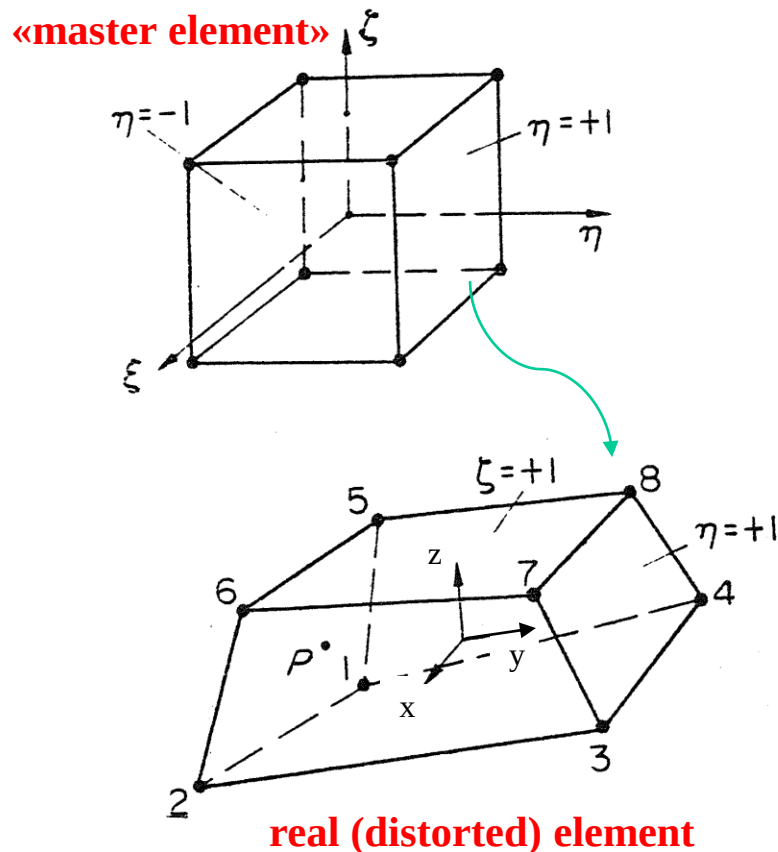
$$\underline{\varepsilon} = [D]\underline{u} = ([D][f])\underline{x}_k = [B]\underline{x}_k$$

$$V_k = \frac{1}{2} \int_{Vol} \underline{\varepsilon}^T \sigma dv = \frac{1}{2} \int_{Vol} \underline{\varepsilon}^T [E] \underline{\varepsilon} dVol = \frac{1}{2} \underline{x}_k^T \underbrace{\int_{Vol} [B]^T [E] [B] dVol}_{[K_k]} \underline{x}_k$$

Assembling of the mass and stiffness matrices follows the same rules as for beam elements

How can this approach be adapted to a general geometry of the solid?

Element distortion needs to be introduced → «mapping»



$$x = x(\xi, \eta, \zeta) = \sum_{i=1}^8 N_i(\xi, \eta, \zeta) x_i$$

$$y = y(\xi, \eta, \zeta) = \sum_{i=1}^8 N_i(\xi, \eta, \zeta) y_i$$

$$z = z(\xi, \eta, \zeta) = \sum_{i=1}^8 N_i(\xi, \eta, \zeta) z_i$$

For $N_i = f_i$ the element is said to be «isoparametric»

Element distortion (I)

Two issues need to be addressed:

- i) The calculation of the stiffness matrix involves the derivatives of the shape functions f_i with respect to the actual space coordinates (x, y, z), not the ones in the master element's space (ξ, η, ζ)
- ii) The infinitesimal volume dv in the actual space must be expressed as a function of $d\xi, d\eta, d\zeta$

$$[M_k] = \int_{Vol} [f]^T \rho [f] dVol$$

$$[K_k] = \int_{Vol} [B]^T [E] [B] dVol$$

Element distortion (II)

$$\frac{\partial f_i}{\partial \xi} = \frac{\partial f_i}{\partial x} \frac{\partial x}{\partial \xi} + \frac{\partial f_i}{\partial y} \frac{\partial y}{\partial \xi} + \frac{\partial f_i}{\partial z} \frac{\partial z}{\partial \xi}$$

$$\frac{\partial f_i}{\partial \eta} = \frac{\partial f_i}{\partial x} \frac{\partial x}{\partial \eta} + \frac{\partial f_i}{\partial y} \frac{\partial y}{\partial \eta} + \frac{\partial f_i}{\partial z} \frac{\partial z}{\partial \eta}$$

$$\frac{\partial f_i}{\partial \zeta} = \frac{\partial f_i}{\partial x} \frac{\partial x}{\partial \zeta} + \frac{\partial f_i}{\partial y} \frac{\partial y}{\partial \zeta} + \frac{\partial f_i}{\partial z} \frac{\partial z}{\partial \zeta}$$

$$\begin{Bmatrix} \frac{\partial f_i}{\partial \xi} \\ \frac{\partial f_i}{\partial \eta} \\ \frac{\partial f_i}{\partial \zeta} \end{Bmatrix} = \underbrace{\begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} & \frac{\partial z}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} & \frac{\partial z}{\partial \eta} \\ \frac{\partial x}{\partial \zeta} & \frac{\partial y}{\partial \zeta} & \frac{\partial z}{\partial \zeta} \end{bmatrix}}_{[J]} \begin{Bmatrix} \frac{\partial f_i}{\partial x} \\ \frac{\partial f_i}{\partial y} \\ \frac{\partial f_i}{\partial z} \end{Bmatrix} \rightarrow \begin{Bmatrix} \frac{\partial f_i}{\partial x} \\ \frac{\partial f_i}{\partial y} \\ \frac{\partial f_i}{\partial z} \end{Bmatrix} = [J]^{-1} \begin{Bmatrix} \frac{\partial f_i}{\partial \xi} \\ \frac{\partial f_i}{\partial \eta} \\ \frac{\partial f_i}{\partial \zeta} \end{Bmatrix}$$

$$dVol = \det([J]) d\xi d\eta d\zeta$$

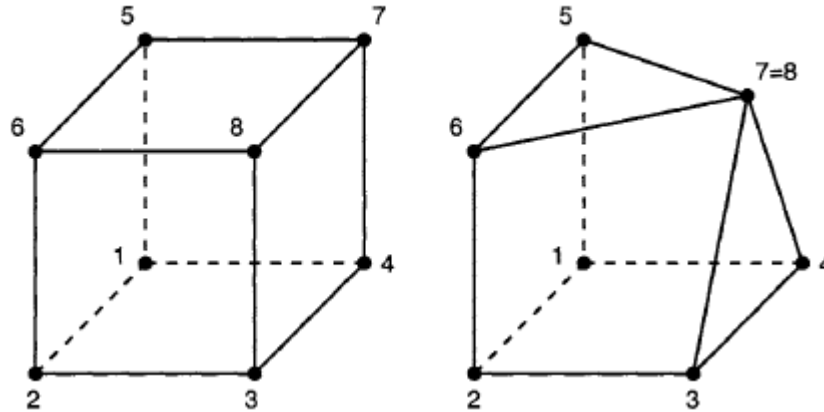
(For a proof, see Appendix F in:

O.C. Zienkiewicz, R.L. Taylor and J.Z. Zhu,

The Finite Element Method Set - Its Basis and Fundamentals

Elsevier, 2005

Effect of excessive distortion



Because (e.g.) 7 and 8 coincide, the mapping between the mater element and real element is not bi-univocal. Hence, $[J]$ becomes singular.

Even if this extreme case is avoided, due to the use of finite algebra the calculaiton of $[J]^{-1}$ becomes ill-conditioned for elements having excessive distorsion, possibly resulting in severe numerical errors in the solution